

Raymond Y. Sheng

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Education:

Cornell University, College of Engineering

Bachelor of Science, Electrical and Computer Engineering

GPA: 3.973/4.300

Ithaca, NY

Expected May 2028

Relevant Courses: Digital Logic & Computer Organization; Embedded Systems; Signals & Systems; Computer Systems Programming; Circuits for Electrical & Computer Engineers; Probability & Inference for Random Signals & Systems

Research and Work Experience:

General Atomics Aeronautical Systems, Inc.

June 2026 - August 2026

Systems Engineering Intern

San Diego, CA

- Pioneered GA company-wide aircraft integration for Army's Next-Generation Command and Control initiative.
- Reimaged and configured a Ground Control Station, installing critical flight monitoring and test software.
- Designed system requirements and verification for US Army Next-Generation Command and Control technology.
- Built AI agents to automate engineering workflows, improving efficiency and evaluating AI model performance.

Cornell University Physical Intelligence

January 2026 - Present

Wireless Power Systems Member

Ithaca, NY

- Developed Wireless Charging Transmitter and Receiver boards to enable hexapod bot to wirelessly charge drone.
- Designed and laid out general-purpose Buck Converter and LDO boards for power regulation testing.
- Assisted development of a diverse portfolio of custom PCBs for Modovolo Drone's UAV platform.
- Built the team as a founding member through early project development and member recruitment.

Cornell Hyperloop

November 2025 - Present

Power Systems Subteam Member

Ithaca, NY

- Designed custom circuits schematic and PCB layout for Emergency Power Off system on hyperloop pod.
- Validated power regulation, protection, and control circuits in SPICE to verify designs prior to PCB fabrication.
- Authored subsystem reports presenting experimental power controls research in Hyperloop technology.
- Presented mid-year reports on electrical subteam progress to a design team of 50+ students and alumni.

Impinj Inc.

May 2024 - August 2024

RF Identification Tag Research Intern

San Diego, CA

- Automated RFID signal measurement across turntable rotations, enabling high-resolution yield data analysis.
- Developed Python and Java data processing scripts to create a framework to accelerate RFID tag data analysis.
- Redesigned turntable and sensing equipment software to streamline RFID tag positioning tests.

Wang Labs, SDSU

June 2021 - December 2023

Wireless Smart Sensing Research Lead

San Diego, CA

- Developed a portable IoT device using Arduino and blockchain-based incentive to promote physical activity.
- Awarded Third Award at Greater San Diego Science Fair in Biomedical Science and Engineering category.
- Led a team of 5+ in discussion and plans in wireless communications research.

FIRST Robotics Team Optix 3749

August 2022 - July 2025

Software Development Engineer

San Diego, CA

- Designed CAD models for a two-DOF robotic arm to transmit flexible movement in rigorous load requirements.
- Programmed autonomous path planning algorithms and vector arm trajectories using inverse kinematics.
- Integrated autonomous motor movement using physics-based motion modeling to optimize path accuracy.

Projects:

Emergency Power Off Board: Designed and validated a safety-critical 2-layer PCB with relay-based emergency cutoff stage in KiCAD, verifying shutdown behavior in SPICE simulation to ensure safe operation of Hyperloop pod.

Wireless Charging Board: Designing inductive coupling transmit/receive coils for hexapod-drone charging in Altium.

Buck Converter and LDO Board: Designed asynchronous buck converter and LDO regulation stages in Altium.

Technical Skills:

Design Tools: Altium, KiCAD, Solidworks, TinkerCAD, Autodesk Inventor, LTspice

Hardware: Analog Circuit Design, FPGA, Arduino, Raspberry Pi, ESP32, UART, RFID Reader, Spectrum Analyzer

Software: Verilog, Claude, Cursor, Docker, Python, Java, C++, C, Linux (Red Hat, Ubuntu), Bash, MATLAB, R, Git